



DEM simulation of small strain and large strain behaviors of granular soils with a coherent contact model

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Abstract

Discrete element method is widely used to study the macroscopic behaviors of granular soils subjected to various loading conditions at a particulate level. However, most of previous studies used different contact models to simulate soil behaviors at different strain levels. To reconcile the disparities between different contact models in DEM simulation, a modified rolling resistance Hertz-Mindlin model is proposed for simultaneously emulating distinct behaviors of granular soils at both small strain and large strain. The Hertz-Mindlin contact model is used to capture the stress-dependent small strain stiffness, while the rolling resistance model is adopted to capture the main effect of particle shape. Contact stiffness degradation is also taken into consideration in the simulation. The comparisons with experimental data show that the model used in this paper can effectively capture the essential features of soil behaviors at both small and large strains in drained and undrained triaxial shear tests. The results indicate that contact stiffness degradation during shearing should be considered for avoiding irrationally high negative excess pore water pressure in undrained triaxial shearing. Incorporation of rolling resistance into the contact model is necessary for reaching a reasonably high critical state stress ratio arising from particle shape.

Keywords DEM · Hertz-Mindlin · Rolling resistance · Contact stiffness degradation · Small strain stiffness · Critical state

1 Introduction

Serviceability limit state design and ultimate limit state design are two conventional design approaches in geotechnical engineering. Stiffness is utilized in a serviceability limit state design, while strength is used in an ultimate limit state design. They are independent in the conventional design. Whereas, their relevance is intrinsic. As shearing proceeds, the soil first behaves linear elastic at very small strain and then gradually approaches the large strain ultimate limit state in a nonlinear manner. Hence, both the small strain and large strain behaviors should be simultaneously considered for completeness.

Discrete element method (DEM) has been well recognized as a suitable approach for studying the macroscopic behaviors of granular soils at a microscopic level [1–8]. However, a unified contact model is rare yet desired to coherently capture the wide spectrum of mechanical properties of granular soil. Most of the previous studies adopted different contact models or inconsistent model parameters to emulate soil behaviors at different strain levels, such as Hertz-Mindlin model and linear model. Hertz-Mindlin model is a widely used nonlinear contact model to obtain the small strain stiffness of granular soil [2, 3, 9–13]. In addition, Hertzian theory was modified to successfully capture the effect of asperity crushing or surface roughness at contacts on the small strain stiffness [14, 15]. Nevertheless, extremely high deviatoric stress develops at large strains in a DEM undrained triaxial test using the same parameters for Hertz-Mindlin model as in small strain stiffness simulations [16]. Some studies have adopted Hertz-Mindlin model with rolling resistance to simulate the behaviors of granular soils at large strains [17, 18], but the adopted particle shear modulus is considerably smaller compared to the one used to obtain the small strain stiffness of granular soil [2, 3, 9–13]. Alternatively, many studies preferred the linear contact

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model or the rolling resistance linear model to emulate large strain behaviors for its simplicity, such as in drained and undrained triaxial simulations [19–23]. Whereas, linear contact model cannot capture the stress dependence of small strain stiffness [15]. More importantly, grain scale tests reveal the obvious nonlinearity of the contact force-displacement relation [24–28]. The shear strength drawn from DEM simulation using linear contact model is generally much lower than those of real sands [4, 21], which is mainly due to the absence of particle shape. Linear contact model with rolling resistance was later developed to better capture the large strain behaviors of granular soils, such as shear strength and shear band formation [6, 20, 29, 30].

The small strain property (e.g., shear wave velocity) has been used to infer the large strain properties (e.g., liquefaction resistance) in engineering practice as well as in DEM simulation, which requires both the small strain and the large strain properties being correctly and simultaneously determined (e.g., Xu et al. [31]). From the perspective of particulate mechanics, the evolution of macroscopic behaviors of granular soils from small strain to large strain is resulted from the interactions of particle contacts. Therefore, a unified contact model is desired in DEM simulation to uniquely capture the macroscopic behaviors of granular soils at both small and large strains.

The explanation for the unrealistically high deviatoric stress in DEM simulation using Hertz-Mindlin model is the neglect of contact stiffness degradation during shearing. Lower contact stiffness will result in lower deviatoric stress in undrained shearing [19, 32]. Figure 1 shows a comparison of tangential force versus displacement between the experimental result in Sandeep et al. [27] and the prediction of Mindlin and Deresiewicz model. The contact stiffness in the experiment decreases with increasing tangential displacement and it is much lower than the prediction of Mindlin and Deresiewicz model because the model is developed for elastic materials and ignores the occurrence of plastic behaviors such as abrasion. Contact stiffness degradation will significantly influence the dilatancy and the deviatoric stress in undrained shearing. The difference between DEM simulation and experiments in undrained triaxial tests of spherical glass beads also provides a strong evidence for the existence of contact stiffness degradation under shearing [16, 33].

To simultaneously emulate the distinct macroscopic behaviors of granular soils at both small and large strains using DEM, a modified Hertz-Mindlin model with rolling resistance is proposed in this work to coherently capture the stress-dependent small stiffness and reasonable large strain shear behaviors. The contact stiffness degradation under shearing is considered as the main modification in this model. Drained and undrained triaxial tests on sand specimens with different densities and under different confining pressures are simulated with the proposed contact model.

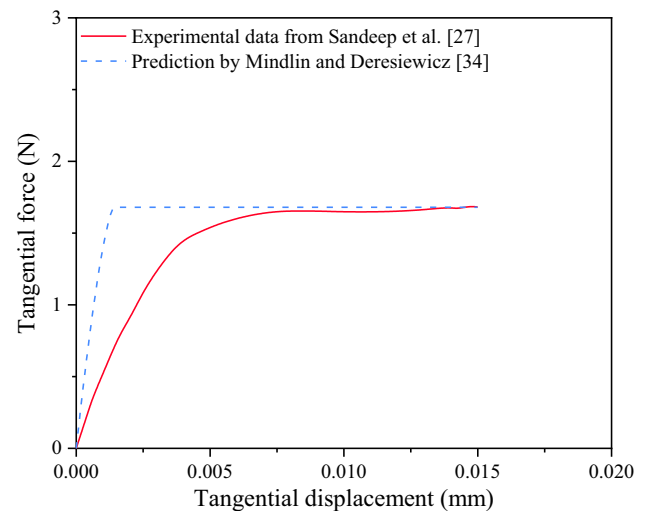


Fig. 1 A comparison of tangential force versus displacement between the experiment data from Sandeep et al. [27] and the prediction by Mindlin and Deresiewicz [34]

The simulated small strain shear stiffness and large strain shear strength are compared with experimental results to verify the performance of the proposed model.

2 Modified Hertz-Mindlin model with rolling resistance

In this section, Hertz-Mindlin model and the rolling resistance at contacts are revisited, which in combination is termed as Hertz-Mindlin model with rolling resistance (rr-Hertz model). Afterwards, contact stiffness degradation is considered during shearing for better emulating the shear behaviors of granular soils, which leads to the modified Hertz-Mindlin model with rolling resistance (mrr-Hertz model).

2.1 Hertz-Mindlin model

The relation between the incremental normal contact force ΔF_n and the incremental depth of indentation Δu between two contacting particles is governed by Hertzian theory [35]

$$\Delta F_n = k_n \Delta u, k_n = \frac{2G_g \sqrt{R^*}}{1 - \nu_g} u^{\frac{1}{2}} \quad (1)$$

where G_g is the shear modulus of the particle, ν_g is the Poisson's ratio of the particle, u is the depth of indentation between two contacting particles, and R^* is the effective contact radius which is defined as $1/R^* = 1/R_1 + 1/R_2$ where R_1 and R_2 are the radii of two contacting particles.

The initial tangential stiffness proposed by Mindlin and Deresiewicz [34] is adopted to govern the relation between the tangential contact force and the tangential displacement [36], which is

$$\Delta F_s = k_s \Delta \delta_s, \quad k_s = \frac{2G_g^{\frac{2}{3}} [6(1-\nu_g)R^*F_n]^{\frac{1}{3}}}{2-\nu_g} \quad (2)$$

where ΔF_s is the incremental tangential contact force and $\Delta \delta_s$ is the incremental relative tangential displacement, respectively. The limiting value of the tangential force F_s^* is governed by Coulomb criterion

$$|F_s^*| = \mu F_n \quad (3)$$

where μ is the inter-particle friction coefficient. The contacting particles slide against each other when the tangential force reaches its limiting value.

2.2 Rolling resistance model

Rolling resistance is adopted to account for particle shapes in this work. The relation between incremental rolling resistance moment ΔM^r and the incremental relative bend-rotation between contacting particles $\Delta \theta_b$ is assumed to be

$$\Delta M^r = -k_r \Delta \theta_b \quad (4)$$

The rolling stiffness k_r is determined with tangential stiffness and effective contact radius as [6]

$$k_r = k_s (R^*)^2 \quad (5)$$

The limiting value of the rolling resistance moment M^* is given as [37]

$$|M^*| = \mu_r R^* F_n \quad (6)$$

where μ_r is the inter-particle rolling friction coefficient.

2.3 Contact stiffness degradation under shearing

Many experimental results indicated that the tangential contact stiffness decreases during shearing due to plastic behaviors such as abrasion [27, 38]. Hence, a two-parameter attenuation function is introduced here to consider the contact stiffness degradation under shearing. The accumulated contact tangential displacement s is chosen as an independent variable according to the behavior of sand grain contacts under tangential force [27] which can be directly recorded in DEM. The effective contact radius R^* is used to normalize the tangential displacement s as to consider the effect of particle size. The tangential contact stiffness under shearing is adjusted by the attenuation function to account for stiffness degradation while the ratio of normal contact stiffness

to the tangential contact stiffness is kept constant suggested by Zhao et al. [39]

$$k'_s = k_s \cdot \left(\frac{1-\alpha}{1 + \left(\frac{s/R^*}{\lambda_r} \right)^4} + \alpha \right) \quad (0 < \alpha < 1) \quad (7)$$

$$k'_n = k'_s \cdot \frac{2-\nu_g}{2(1-\nu_g)} \quad (8)$$

where k'_n , k'_s are the degraded normal and tangential contact stiffness during shearing, k_s is the initial tangential contact stiffness determined by Eq. (2), λ_r is a reference strain which controls the characteristics of contact stiffness degradation, α is a coefficient determining the residual value of contact stiffness which represents the degree of deterioration caused by shearing. Note that the rolling stiffness k_r is also updated using the degraded tangential contact stiffness k'_s according to Eq. (5).

The results of particle contact level tests are used to verify the performance of the proposed attenuation function. Figure 2 compares the tangential contact stiffness coefficient k'_s/k_s obtained by Sandeep and Senetakis [38] and the fitting curves of Eq. (7). The result shows that the proposed attenuation function can well capture the contact stiffness degradation. As seen in Fig. 2, the tangential contact stiffness starts to decrease sharply when s/R^* exceeds about 4×10^{-4} and it approaches the residual value when s/R^* exceeds about 1×10^{-2} . It seems that a reference strain λ_r of 2.0×10^{-3} can capture the main feature of the experimental data and it is adopted in the following simulation without specification. Meanwhile, the residual contact stiffness coefficient α seem

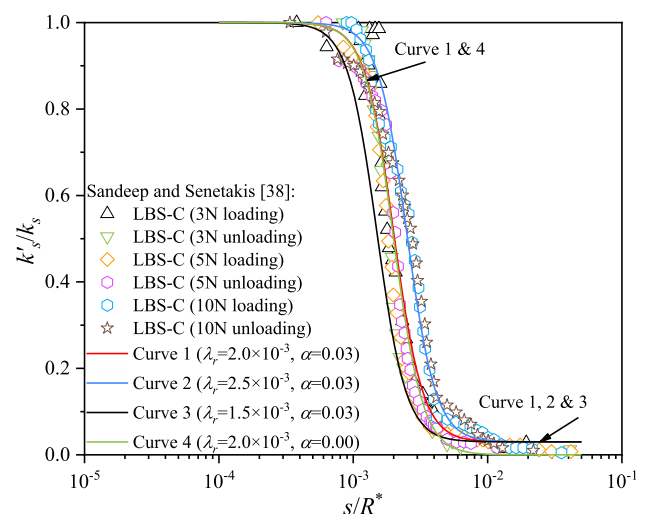


Fig. 2 A comparison of the contact stiffness degradation between experiments and the fitting results using Eq. (7)

to be in the range of 0–0.03 and a value of 0.03 is adopted in the following simulation without specification. By contrast, if the contact stiffness degradation is not considered in the model, the contact stiffness is constant before slip which is not consistent with the experimental data. Moreover, parametric studies will also be carried out to investigate the parameters of λ_r and α on the mechanical behaviors of granular soils in the following.

3 DEM simulation

The DEM simulations were performed with the well-recognized DEM software PFC3D. The contact model was implemented via C++ plugin.

3.1 Specimen generation

The specimen was a $4.5 \times 4.5 \times 9$ mm³ cuboid surrounded by six rigid walls and it contains 44,488 spherical particles. Figure 3 shows the particle size distribution of the particle assembly, which is in accordance with those of real Toyoura sand [40–42].

The specimen was generated by radius expansion method. First, particles were generated and cycled to reach an equilibrium state. During these procedures, the friction coefficient μ and rolling resistance coefficient μ_r (if any) were set to zero. Then, the specimen was subjected to a confining stress of 10 kPa and cycled to reach equilibrium. At this initial stage, different inter-particle friction coefficients were used to generate specimens of different void ratios [43]. Generally, the maximum void ratio was obtained using the same friction coefficient and rolling resistance coefficient (if

any) at shearing, while the minimum void ratio was obtained using zero friction coefficient and rolling resistance coefficient (if any). The friction coefficient μ and rolling resistance coefficient μ_r were subsequently set to designated values to represent the soil particle characteristics in further consolidation and shearing. Note that the contact stiffness degradation is not activated before shearing in the simulations.

3.2 Small strain drained triaxial test

Small strain triaxial test was carried out to measure the small strain shear stiffness [9]. The shearing was terminated when the probing shear strain reached 10^{-6} . The deformation of the specimen at this stage mainly results from the inter-particle displacements with the microstructure preserved [44]. The small strain shear stiffness G_0 is determined by

$$G_0 = \frac{\Delta\tau}{\Delta\gamma} = \frac{\Delta\sigma_v - \Delta\sigma_h}{2(\Delta\varepsilon_v - \Delta\varepsilon_h)} \quad (9)$$

where $\Delta\tau$ is the incremental shear stress, $\Delta\gamma$ is the incremental shear strain, $\Delta\sigma_v$ is the incremental axial stress, $\Delta\sigma_h$ is the incremental radial stress, $\Delta\varepsilon_v$ is the incremental axial strain, and $\Delta\varepsilon_h$ is the incremental radial strain.

3.3 Large strain triaxial tests

Both drained and undrained triaxial tests were simulated. The mean effective stress p' , deviatoric stress q and pore pressure Δu are defined as

$$p' = (\sigma'_v + 2\sigma'_h)/3 \quad (10)$$

$$q = \sigma'_v - \sigma'_h \quad (11)$$

$$\Delta u = p'_0 - p' + \frac{q}{3} \quad (12)$$

where σ'_v and σ'_h are the average effective normal stresses in the vertical and horizontal directions respectively, and p'_0 is the initial mean effective stress after isotropic consolidation.

In drained triaxial simulations, the top and bottom walls moved vertically at the same velocity while the confining pressure was maintained. In undrained simulations, each wall moved at the intended velocities to ensure the constant volume [32, 45–47]. The axial strain rate is 2 s^{-1} for both drained and undrained shearing tests. A timestep of $1 \times 10^{-8} \text{ s}$ was adopted based on the particle mass, moment of inertia and contact stiffness [48]. The maximal inertial number I remained smaller than 0.003 throughout the drained and undrained shearing tests for achieving a quasi-static state [49–52].

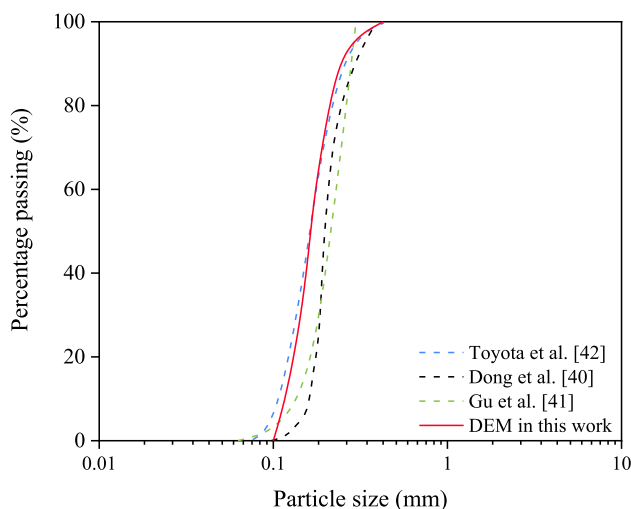


Fig. 3 Particle size distribution of Toyoura sand in DEM simulation and experiments

3.4 Parameters determination

Simulations with four different contact models were conducted for better illustration. The physical parameters are listed in Table 1. The prefix ‘rr’ stands for rolling resistance. For example, ‘mrr-Hertz’ denotes modified Hertz-Mindlin model with rolling resistance. Hertz-Mindlin model (Hertz), Hertz-Mindlin model with rolling resistance (rr-Hertz), and linear model with rolling resistance (rr-Linear) were also adopted in the simulation for comparison. The particle density was 2640 kg/m^3 in all simulations.

In rolling resistance Hertz-Mindlin model, the particle Poisson’s ratio ν_g was set to 0.15 as suggested by Gong et al. [2] and Gu et al. [43]. A particle shear modulus G_g of 23 GPa was adopted which was calibrated according to the G_0 value of a Toyoura sand specimen with a void ratio of 0.726 at 100 kPa. Note that this particle shear modulus is also consistent with the measured sand particle shear modulus in Sandeep and Senetakis [38]. The inter-particle friction coefficient μ and rolling resistance coefficient μ_r were set in accordance with Gu et al. [20] to emulate the realistic void ratio and critical state stress ratio. In the modified rolling resistance Hertz-Mindlin model, the reference strain λ_r is set to be 2.0×10^{-3} based on the particle contact level tests and the residual contact stiffness coefficient α is set to be 0.03.

For better illustration, another two particle shear moduli were adopted in Hertz-Mindlin contact model for small strain and large strain shearing tests. The particle shear modulus G_g of 23 GPa was adopted for simulations at small strain, while a particle shear modulus G_g of 1.46 GPa was

adopted for simulations at large strains [32]. The same model parameters were adopted in the linear model with rolling resistance as those in Gu et al. [20] at both small and large strains for comparison. Tables 2 and 3 present the complete simulation program at small strain and large strain, respectively.

4 Results and discussions

4.1 Small strain shear stiffness G_0

Figure 4 shows the stress–strain curves of specimens with almost the same void ratio at a confining pressure of 100 kPa with different contact models in small strain drained triaxial tests. Note that the specimen labels sequentially consist of contact model, contact model, initial confining stress, and relative density after consolidation. For example, mrr-Hertz-100-67.2% refers to the simulation using the mrr-Hertz model with a specimen at an initial relative density 67.2% under a confining pressure of 100 kPa. Each curve increases in a linear manner at small strain satisfying the linear elastic assumption. The small strain shear modulus G_0 is calculated by Eq. (9).

Figure 5 shows the variations of G_0 with void ratio e at a confining pressure of 100 kPa in simulations with different contact models, together with the experimental data of Toyoura sand in Youn et al. [53] and Gu et al. [41]. The value of G_0 decreases as void ratio increases (or density decreases), as expected. The existence of rolling resistance increases the G_0 value, while the contact stiffness degradation does not affect the G_0 value. In addition, the values of G_0 obtained by rr-Linear model are significantly lower than those obtained by other non-linear contact models. The relative deviation is about 25% between experimental data and DEM simulation results using non-linear contact models. It should be noted that the contact stiffness and therefore the G_0 value of a specimen are proportional to $(G_g)^{2/3}$ as indicated in Eq. (2), which means that different G_0 values will be obtained for different G_g values adopted in DEM simulations. The G_0 values obtained by non-linear contact models decrease faster with increasing void ratio than the experimental data does. It is probably caused by the difference of void ratio attained in DEM simulation with different contact models and that in reality. The void ratio range generated by Hertz model (i.e., 0.562–0.696) is much narrower than that by rr-Hertz and mrr-Hertz model (i.e., 0.560–0.810), and they are both narrower than those in experiments (i.e., 0.605–0.977). From this point of view, irregular particles (i.e., clumps, ellipsoids, polyhedrons) should be used in future to achieve a more realistic void ratio range.

In the past decades, enormous studies have been carried out on the small strain shear modulus G_0 . It is well

Table 1 Input DEM model parameters in this paper

Contact models	Model parameters	Value
rr-Linear	Particle normal stiffness k_n^*	$k_n = k_0 \cdot r$
	Particle shear stiffness k_s^*	$k_s = \alpha \cdot k_n$
	Inter-particle friction coefficient μ	0.45
	Rolling resistance coefficient μ_r	0.50
Hertz	Particle shear modulus (small strain) G_g	23.0 GPa
	Particle shear modulus (large strain) G_g	1.46 GPa
	Particle Poisson’s ratio ν_g	0.15
	Inter-particle friction coefficient μ	0.50
rr-Hertz/mrr-Hertz	Particle shear modulus G_g	23.0 GPa
	Particle Poisson’s ratio ν_g	0.15
	Inter-particle friction coefficient μ	0.45
	Rolling resistance coefficient μ_r	0.5
	[†] Reference strain λ_r	2.0×10^{-3}
	[†] Residual contact stiffness coefficient α	0.03

* $k_0 = 8 \times 10^8 \text{ N/m}^2$, $\alpha = 0.15$ and r is the radius of the particle; [†]These model parameters are only used in mrr-Hertz model

Table 2 Simulation program at small strains

Contact model ($e_{\max}$, $e_{\min}$)	Void ratio/relative density	Confining pressure (kPa)	Contact model ($e_{\max}$, $e_{\min}$)	Void ratio/Relative density	Confining pressure (kPa)
rr-Linear ($e_{\max} = 0.839$, $e_{\min} = 0.555$)	0.734/37.0%	50	rr -Hertz ($e_{\max} = 0.810$, $e_{\min} = 0.560$)	0.727/33.2%	50
	0.731/38.0%	100		0.726/33.6%	100
	0.727/39.4%	200		0.724/34.4%	200
	0.718/42.6%	400		0.721/35.6%	400
	0.689/52.8%	50		0.679/52.4%	50
	0.687/53.5%	100		0.678/52.8%	100
	0.683/54.9%	200		0.677/53.2%	200
	0.676/57.4%	400		0.675/54.0%	400
	0.651/66.2%	50		0.643/66.8%	50
	0.649/66.9%	100		0.642/67.2%	100
	0.646/68.0%	200		0.641/67.6%	200
	0.640/70.1%	400		0.639/68.4%	400
Hertz ($e_{\max} = 0.696$, $e_{\min} = 0.562$)	0.650/34.3%	50	mrr -Hertz ($e_{\max} = 0.810$, $e_{\min} = 0.560$)	0.727/33.2%	50
	0.649/35.1%	100		0.726/33.6%	100
	0.648/35.8%	200		0.724/34.4%	200
	0.647/36.6%	400		0.721/35.6%	400
	0.630/49.3%	50		0.679/52.4%	50
	0.629/50.0%	100		0.678/52.8%	100
	0.629/50.0%	200		0.677/53.2%	200
	0.627/51.5%	400		0.675/54.0%	400
	0.608/65.7%	50		0.643/66.8%	50
	0.608/65.7%	100		0.642/67.2%	100
	0.607/66.4%	200		0.641/67.6%	200
	0.606/67.2%	400		0.639/68.4%	400

Table 3 Simulation program at large strains

Contact model	Void ratio/relative density	Confining pressure (kPa)	Test type	Void ratio/relative density	Confining pressure (kPa)	Test type
rr-Linear	0.726/39.8%	100	CD	0.726/39.8%	100	CU
	0.713/44.4%	400	CD	0.732/37.7%	1000	CU
	0.608/81.3%	100	CD	0.658/63.7%	100	CU
	0.600/84.2%	400	CD	0.657/64.1%	1000	CU
Hertz	0.647/36.6%	100	CD	0.647/36.6%	100	CU
	0.647/36.6%	400	CD	0.644/38.8%	1000	CU
	0.585/82.8%	100	CD	0.612/62.6%	100	CU
	0.585/82.8%	400	CD	0.612/62.6%	1000	CU
rr-Hertz	0.715/38.0%	100	CD	0.715/38.0%	100	CU
	0.710/40.0%	400	CD	0.712/39.2%	1000	CU
	0.621/75.6%	100	CD	0.653/62.8%	100	CU
	0.604/82.4%	400	CD	0.654/62.4%	1000	CU
mrr-Hertz	0.715/38.0%	100	CD	0.715/38.0%	100	CU
	0.710/40.0%	400	CD	0.712/39.2%	1000	CU
	0.621/75.6%	100	CD	0.653/62.8%	100	CU
	0.604/82.4%	400	CD	0.654/62.4%	1000	CU

CD consolidated drained triaxial test, CU consolidated undrained triaxial test

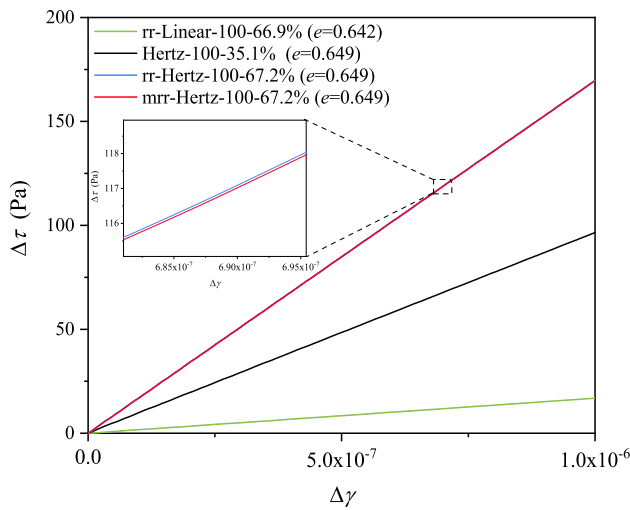


Fig. 4 Stress–strain curves of specimens at $p'_0 = 100$ kPa using different contact models

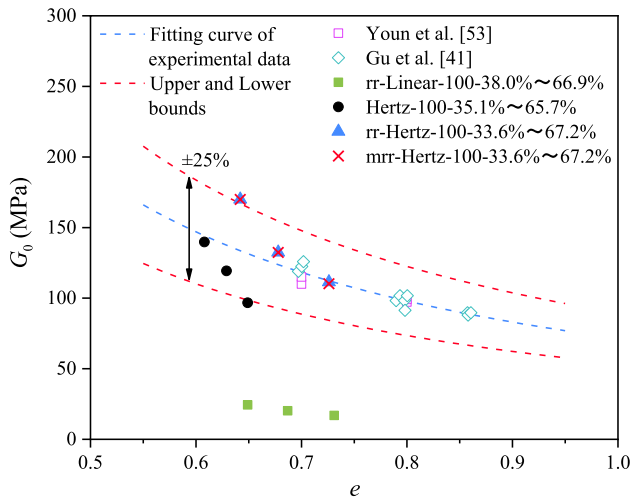


Fig. 5 Variation of G_0 with e at $p'_0 = 100$ kPa in DEM simulations and experiments

accepted that the G_0 of granular soils is a function of the mean effective stress p'_0 and void ratio e , and it can be expressed in a power function form as [54]

$$G_0 = AF(e) \left(\frac{p'_0}{p_a} \right)^n \quad (13)$$

where A is a constant reflecting soil type, grain properties and fabric, p_a is a reference stress (98 kPa in this study), n is the stress exponent reflecting the effect of confining pressure and $F(e)$ is a void ratio function reflecting the effect of soil density. Hardin and Richart [54] proposed a widely used form of $F(e)$ as

$$F(e) = \frac{(b - e)^2}{(1 + e)} \quad (14)$$

where b is an intrinsic property that depends on the particle shape (i.e., b equals 2.17 for rounded particles and 2.97 for angular ones).

Figure 6 shows the evolution of G_0 with p'_0/p_a using different contact models for specimens with similar void ratios (e.g., in the range of 0.639–0.651), together with the experimental results from Gu et al. [41]. The G_0 value increases as p'_0 increases as expected and the fitting curve using Eq. (13) is also shown in the figure. It also indicates that the G_0 obtained from rr-Hertz model and mrr-Hertz model is larger than that obtained from Hertz model due to the presence of rolling resistance. The G_0 values obtained from rr-Hertz model and mrr-Hertz model are almost the same. The stress exponents n obtained from Hertz, rr-Hertz, and mrr-Hertz models are 0.392, 0.363, and 0.344 respectively, which are comparable to the experimental value of 0.41 [41]. Nevertheless, the stress exponent n of rr-Linear model is 0.094 which is much smaller than the experimental data, indicating that G_0 is almost independent of confining pressure. It is clear that the linear contact model cannot successfully capture the stress-dependent characteristic of small strain shear stiffness [15].

Figure 7 plots the G_0 values normalized by the void ratio function $F(e)$ against the normalized pressure in DEM simulations and experiments [41, 53, 55–57]. As seen in Fig. 7a, the simulation results using rr-Hertz model and mrr-Hertz model are comparable with experimental data. It can be also seen that the rr-Linear model cannot reflect the stress dependence of G_0 , since the contact stiffness is constant which is far from the reality. It is reasoned that the slight

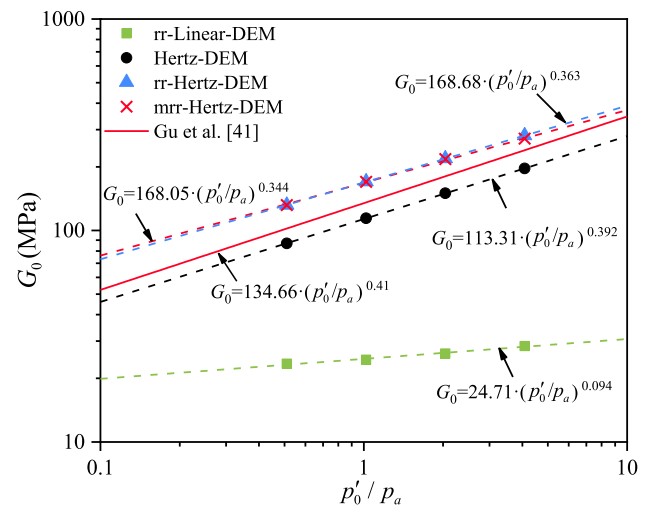


Fig. 6 Relationship between G_0 and p'_0/p_a of specimens at similar void ratios

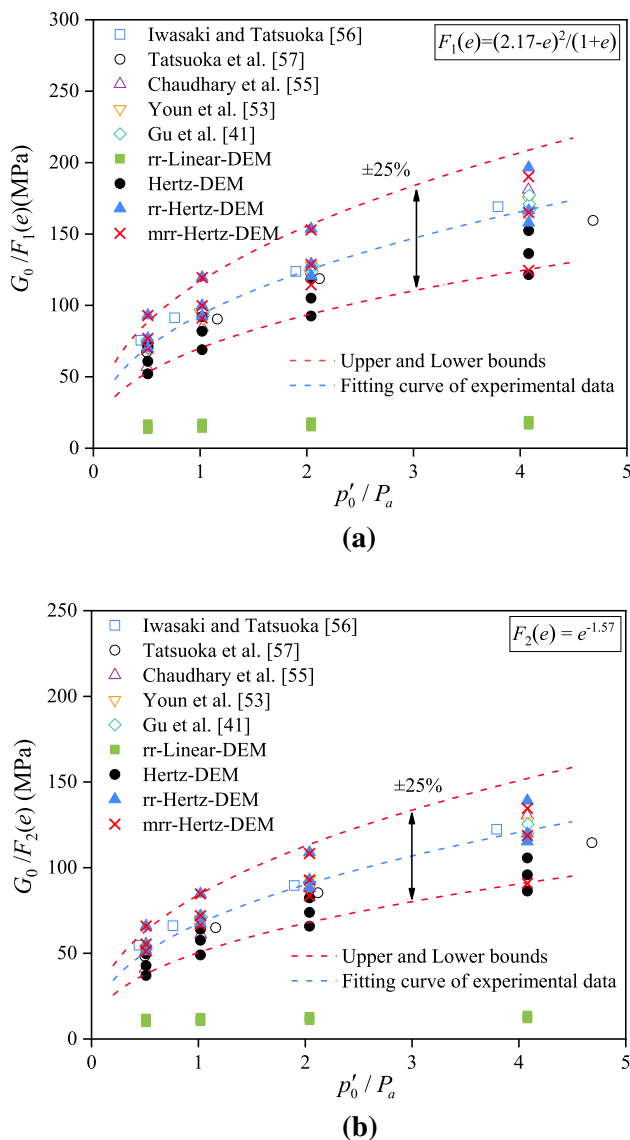


Fig. 7 Comparison of G_0 of Toyoura sand normalized by void ratio functions: **a** $F_1(e) = (2.17 - e)^2 / (1 + e)$; **b** $F_2(e) = e^{-1.57}$ in DEM simulation and in experiments

increase of G_0 with confining pressure in Fig. 7 for rr-linear contact model is resulted from the decrease of void ratio or increase of coordinate number with increasing confining pressure [11]. Another common void function $F(e) = e^{-x}$ proposed by Lo Presti et al. [58] was adopted as well for comparison, as shown in Fig. 7b. This form can be justified with Hertzian contact theory for perfect spheres by considering an idealized assembly of equal sized spheres [59]. In this study, an exponent value of 1.57 was used which is an average value derived from the experiment results in Gu et al. [41]. Figure 7 shows that the deviation of the simulation results using Hertz, rr-Hertz, and mrr-Hertz model is about 25% from the experiment results which demonstrates

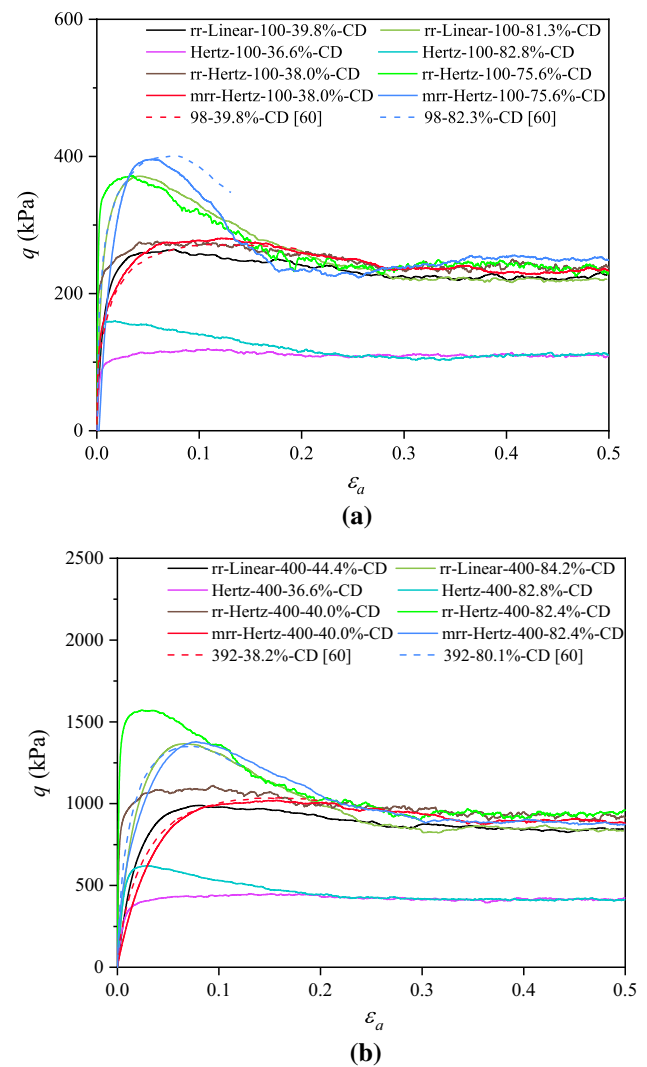


Fig. 8 Deviatoric stress against axial strain in drained shearing using different contact models at **a** 100 kPa and **b** 400 kPa

the effectiveness of the proposed model for capturing the small strain stiffness behavior.

4.2 Macroscopic behaviors at large strains

Figure 8 shows the variation of deviatoric stress q against axial strain ϵ_a in drained triaxial tests on specimens with different densities using different contact models at 100 kPa and 400 kPa, together with the experimental results from Fukushima and Tatsuoka [60]. Note that the specimen labels sequentially consist of contact model, initial confining pressure, relative density after consolidation and drainage condition during shearing. The deviatoric stress increases first and then decreases to constant values at about 30% axial strain. Identical deviatoric stresses are achieved using the same contact model at the same confining pressure with different relative densities, which is consistent with the critical

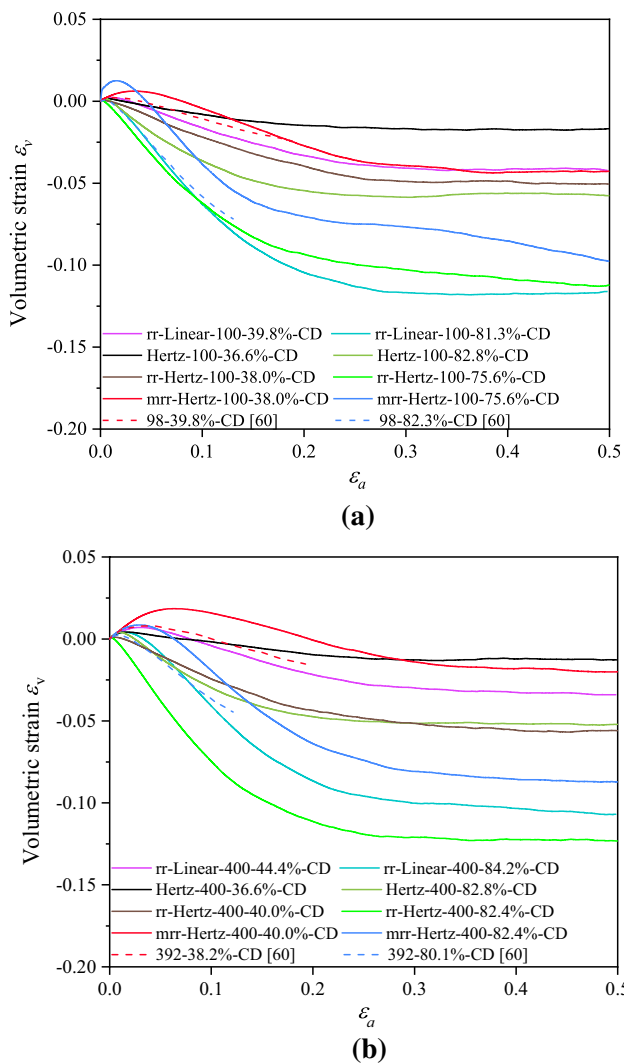


Fig. 9 Volumetric strain against axial strain in drained shearing using different contact models at **a** 100 kPa and **b** 400 kPa

state theory. The results indicate that the rolling resistance can significantly increase the shear strength [6] while the consideration of contact stiffness degradation decreases the peak drained strength at a high confining pressure. It also indicates that the influence of contact stiffness degradation differs at different confining pressures. Simulations with all contact models except for Hertz model and rr-Hertz model agree well with the experimental data. The specimens with Hertz model manifests much smaller drained shear strength while the specimens using rr-Hertz model shows a much higher drained peak strength at 400 kPa.

Figure 9 plots the volumetric strain ε_v against the axial strain ε_a in drained triaxial shearing tests using different contact models at 100 kPa and 400 kPa, together with the experimental results from Fukushima and Tatsuoka [60]. All specimens manifest a contractive behavior first and then a dilative behavior. When approaching the critical state, the

volumetric strain seems to be relatively constant. Moreover, higher confining pressure or lower relative density leads to a more contractive behavior, which is consistent with the experiment data. Meanwhile, the rolling resistance increases the dilatancy, while the consideration of contact stiffness degradation significantly decreases the dilatancy. Generally, the results of rr-Linear model match the experimental results well at both 100 kPa and 400 kPa, while the dilatancy of the specimens using mrr-Hertz model is somewhat lower than the experiments.

Figure 10 compares the variations of deviatoric stress against axial strain in undrained shearing tests between simulations using different contact models and experimental data in Verdugo and Ishihara [61]. The deviatoric stress of each specimen increases with the increase of axial strain, as expected. The critical state is reached at an axial strain of about 30%. It can be found that the deviatoric stress obtained from simulations using mrr-Hertz, Hertz model and rr-Linear model are all comparable with the experimental data. Extremely high deviatoric stress is achieved using rr-Hertz model or Hertz model with a high particle shear modulus [32]. The consideration of contact stiffness degradation results in a much smaller and reasonable deviatoric stress. Note that the particle shear modulus G_g was set to 1.46 GPa in Hertz model to reduce the dilatancy and achieve a reasonable peak deviatoric stress as suggested by Keishing and Hanley [32]. However, a coherent set of model parameters are adopted using mrr-Hertz model, which is more reasonable. Rolling resistance linear contact model does have better performance than other two contact models, but the results obtained from mrr-Hertz and Hertz model are also acceptable [32].

Figure 11 shows the evolution of excess pore water pressure Δu in the numerical undrained triaxial tests, together with the experimental data in Verdugo and Ishihara [61]. Δu increases first and then decreases in all specimens, which reflects the evolution of dilatancy of the specimens. It also shows that higher density and lower confining pressure increase the magnitude of negative excess pore water pressure as expected. Moreover, DEM simulations using mrr-Hertz or rr-Linear model produce acceptable evolution of Δu compared to the experiment results. However, Δu from simulations using rr-Hertz model is significantly higher than that in the experiments, which indicates the importance of the contact stiffness degradation.

Figure 12 shows the stress paths in q - p' space from both drained and undrained triaxial tests using different contact models. As seen in Fig. 12, the critical stress ratio M obtained from mrr-Hertz is 1.24 which is consistent with the experimental value 1.24 [61], while much smaller value 0.77 is obtained from Hertz model and a slightly higher value 1.32 is obtained from rr-Hertz model. It indicates that the rolling resistance can significantly increase the critical stress

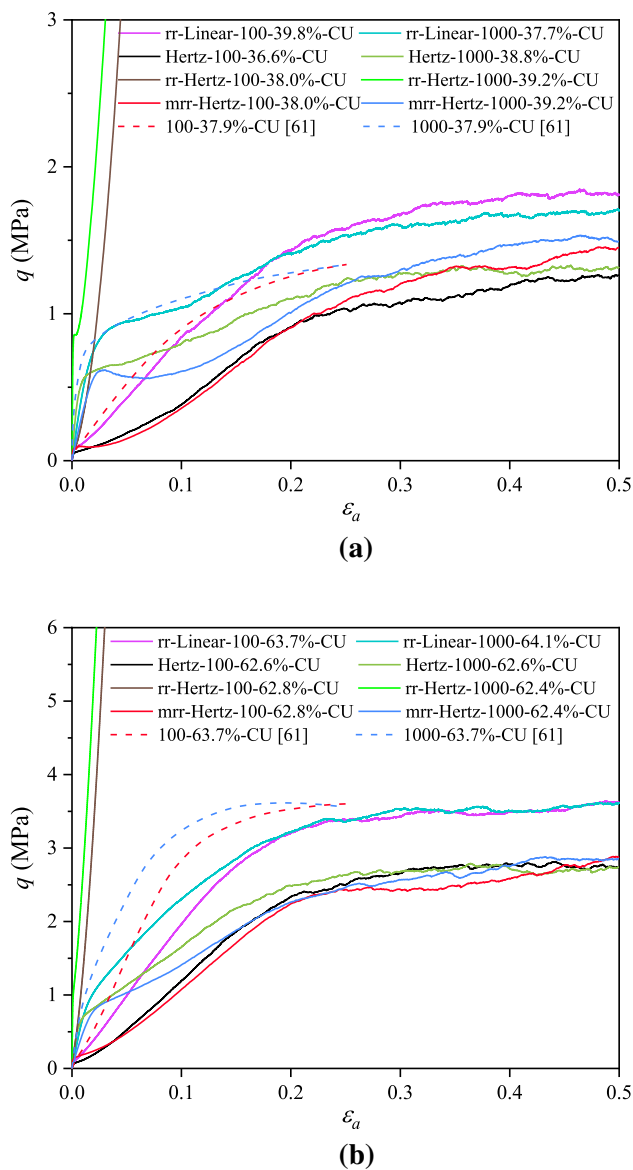


Fig. 10 Deviatoric stress against axial strain in undrained triaxial tests at different confining pressures using different contact model on **a** medium dense and **b** dense specimen

ratio M , while the consideration of contact stiffness degradation decreases the critical stress ratio. Note that the simulation using Hertz model can generate reasonable deviatoric stress compared to the experimental results (see Fig. 10a, b) due to the high negative excess pore water pressure (see Fig. 11a, b), although the simulated critical state stress ratio is much lower than the experimental ones. The critical stress ratio M using rr-linear model is 1.16 which can be deemed as reasonable [20], but the small strain stiffness is independent of stress that is opposite to the experiments. Overall, it can be concluded that mrr-Hertz model is able to simultaneously and successfully emulate the behaviors of granular soil at

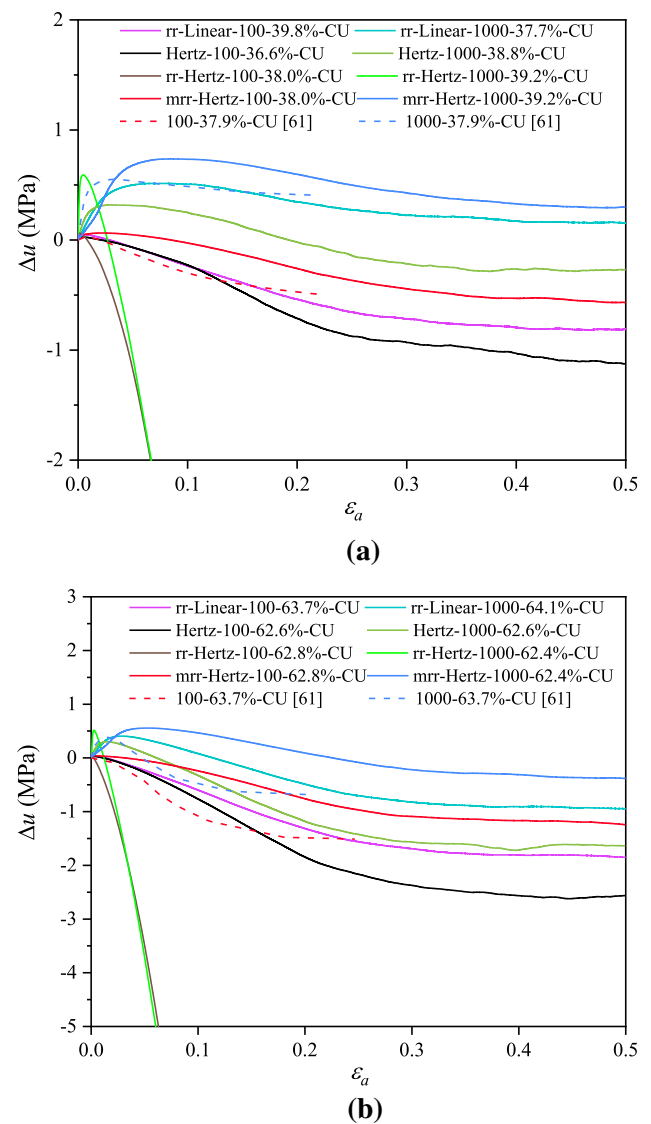


Fig. 11 Excess pore water pressure in undrained triaxial tests at different confining pressures using different contact models on **a** medium dense and **b** dense specimens

both small and large strains given the results in Figs. 7, 8, 10 and 12.

4.3 Parametric study

A parametric study is carried out to explore the influence of reference strain λ_r and residual contact stiffness coefficient α on drained and undrained behaviors at large strains. According to experimental results shown in Fig. 2, λ_r varies from 1.5×10^{-3} to 2.5×10^{-3} , while α varies from 0.02 to 0.04, respectively. Figures 13 and 14 show the influences of λ_r and α on the deviatoric stress and volumetric strain under drained shearing for a specimen of 38% relative density at 100 kPa. It indicates that λ_r has a negligible effect on the deviatoric stress and volumetric

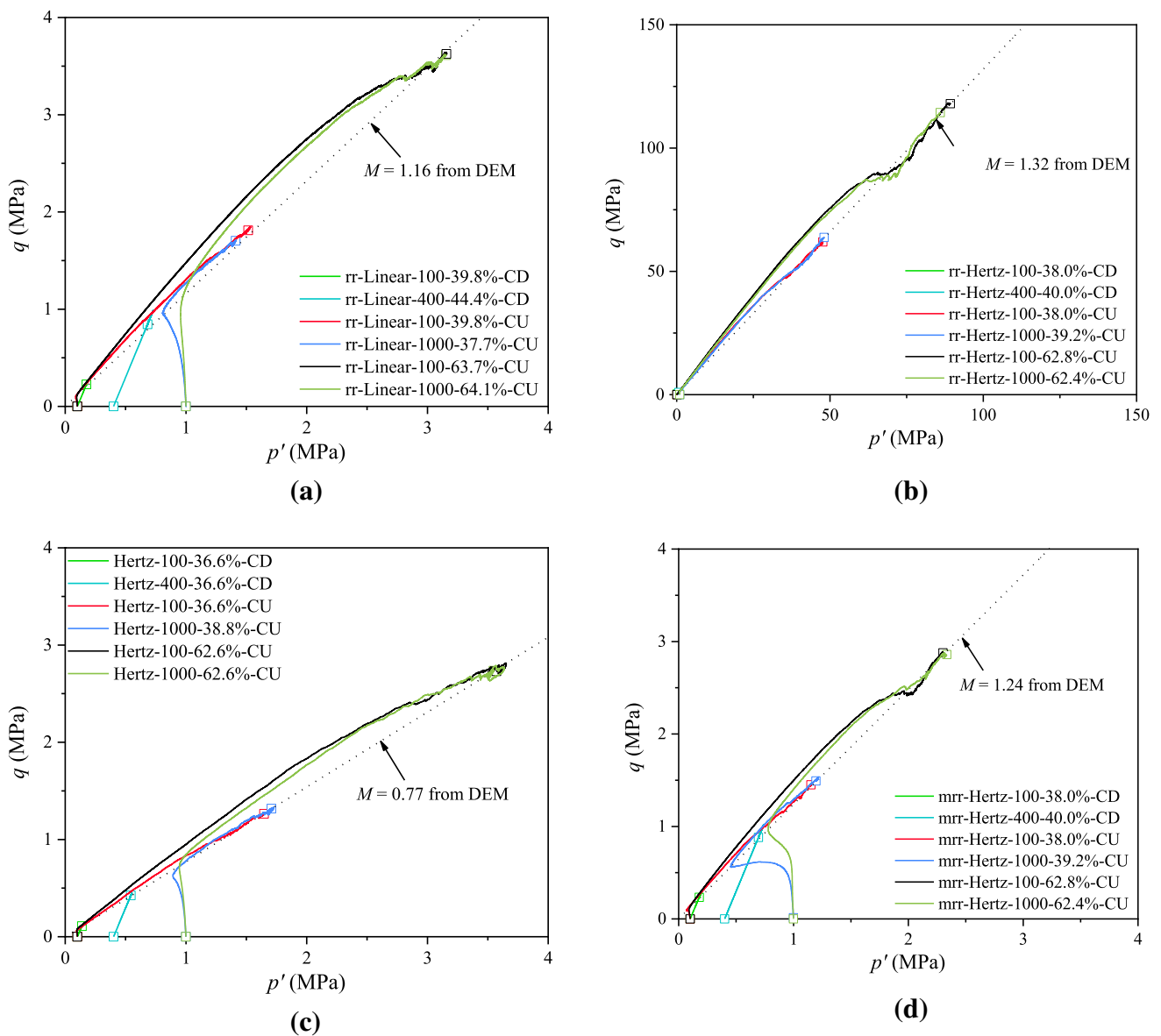


Fig. 12 Stress paths and critical state lines in q - p' space using **a** rr-linear model, **b** Hertz model, **c** rr-Hertz model, and **d** mrr-Hertz model

strain in a range from 1.5×10^{-3} to 2.5×10^{-3} . Meanwhile, α also has a negligible effect on the deviatoric stress, while higher α contributes to higher dilatancy. Figures 15 and 16 shows the influences of λ_r and α on the deviatoric stress and excess pore water pressure under undrained shearing using the same specimen in drained shearing. Similarly, λ_r has negligible effects on the deviatoric stress and excess pore water pressure under undrained shearing, while higher α contributes to higher deviatoric stress and higher negative excess pore water pressure. In conclusion, the drained and undrained behaviors are not sensitive to the reference strain λ_r , while the residual contact stiffness coefficient α contribute to higher dilatancy. Comparing the simulation results and the experimental data, the reference strain λ_r of 2.0×10^{-3} and residual contact stiffness

coefficient α of 0.03 are used in the previous study. Note that there are still some differences between the simulation and experiment (see Figs. 14, 15, 16), especially in undrained shearing. Further studies are necessary to further improve the performance of the proposed contact model.

5 Conclusion

A modified Hertz-Mindlin model with rolling resistance for simultaneously emulating the mechanical behavior of granular soils at both small and large strains is proposed in this work. The contact stiffness degradation during shearing is considered in the model. Numerical drained and undrained

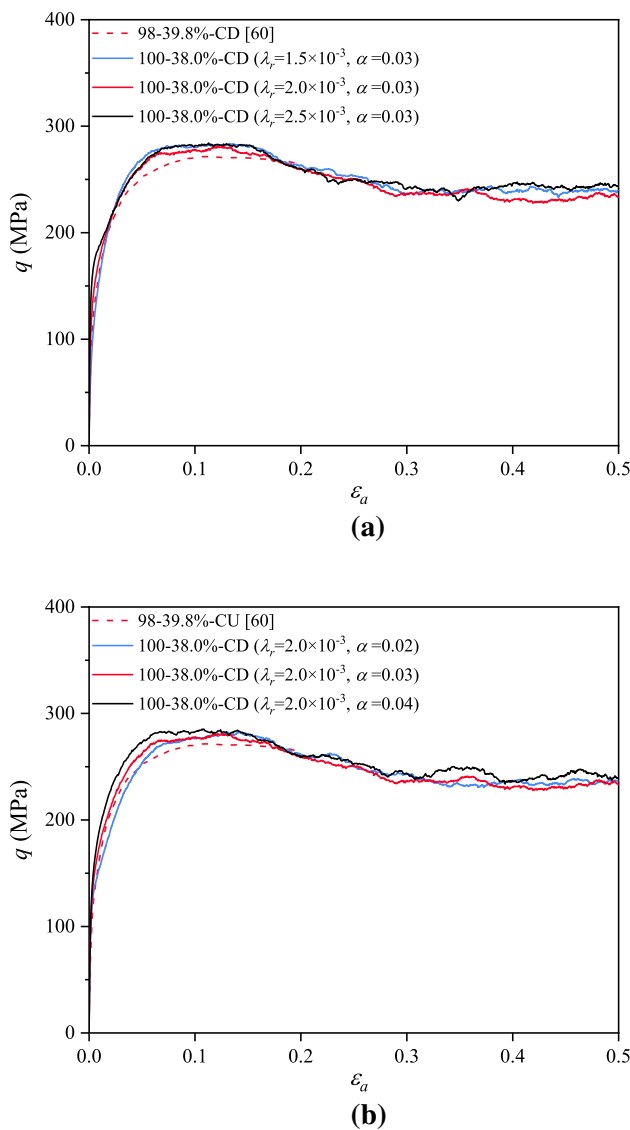


Fig. 13 Influences of **a** λ_r and **b** α on the deviatoric stress under drained shearing

triaxial tests were conducted to investigate the small strain shear stiffness and shear strength of granular soils. The major findings are summarized as follows:

- (1) The modified Hertz-Mindlin model with rolling resistance can effectively characterize the stiffness degradation response between particles during shearing at a particulate level and reasonably emulate the realistic mechanical behaviors of granular soil at both small and large strains. The linear contact model fails to simulate the stress-dependent stiffness at small strain, while the rolling resistance can increase the small strain shear stiffness.
- (2) The variations between the small strain stiffness and void ratio in DEM simulation is more significant com-

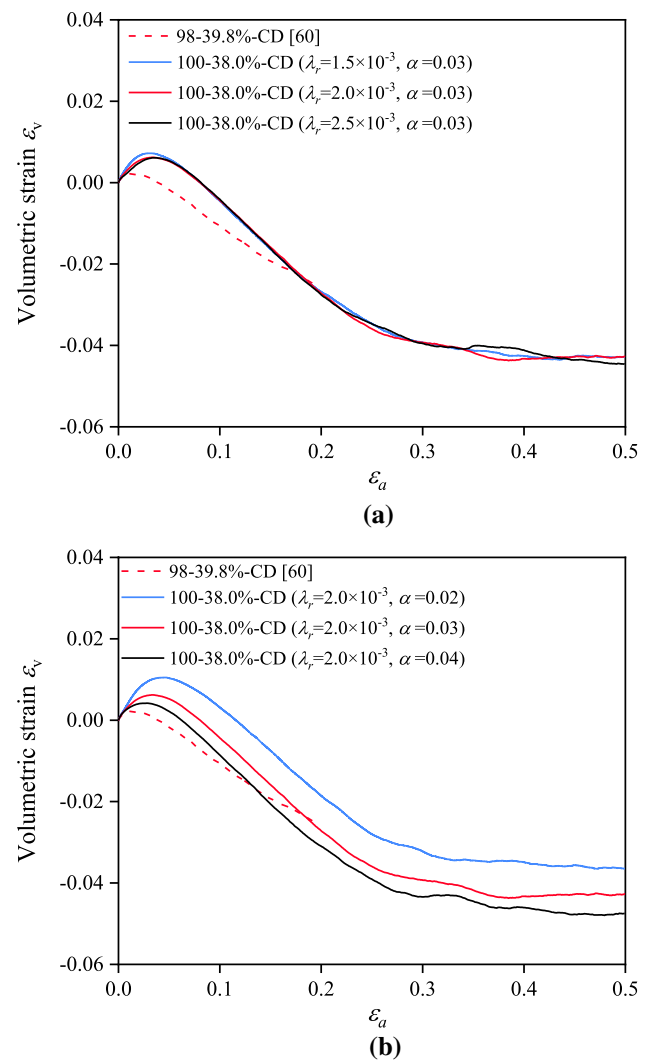


Fig. 14 Influences of **a** λ_r and **b** α on the volumetric strain under drained shearing

pared to the experiment results, especially for the Hertz model, which is probably due the small range of void ratio can be achieved in the DEM simulation. It is suggested to adopt more realistic particle shape to generate a wider void ratio range in future simulations.

- (3) The Hertz contact model cannot yield realistic shear stress ratio and extremely high excess pore pressure is generated using rr-Hertz model. Deviatoric stress decreases due to the consideration of contact stiffness degradation which is more reasonable considering the experimental results. It is suggested that both drained and undrained shearing behavior should be benchmarked against the experimental results to achieve an approving contact model in DEM.
- (4) Although the modified Hertz-Mindlin model with rolling resistance have a better performance than the present contact models, there are still some differences

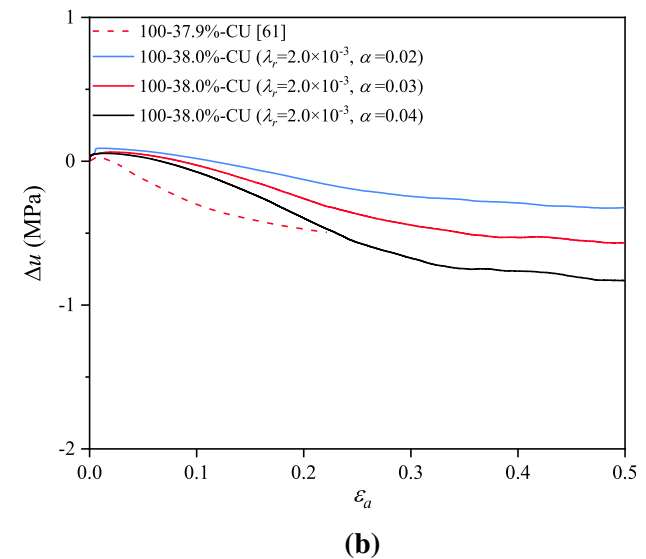
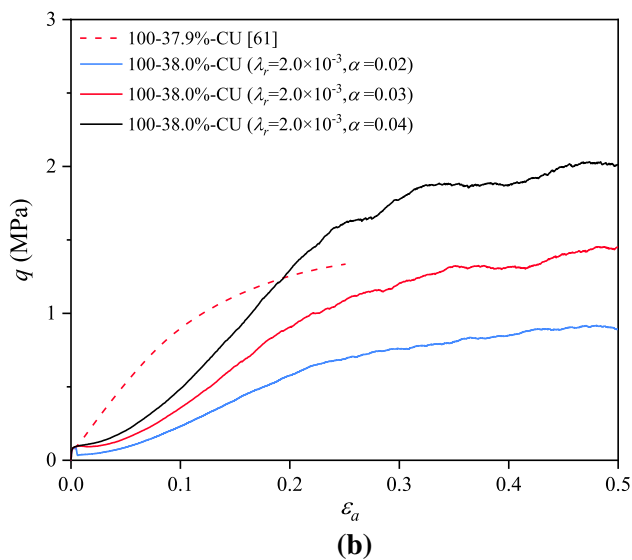
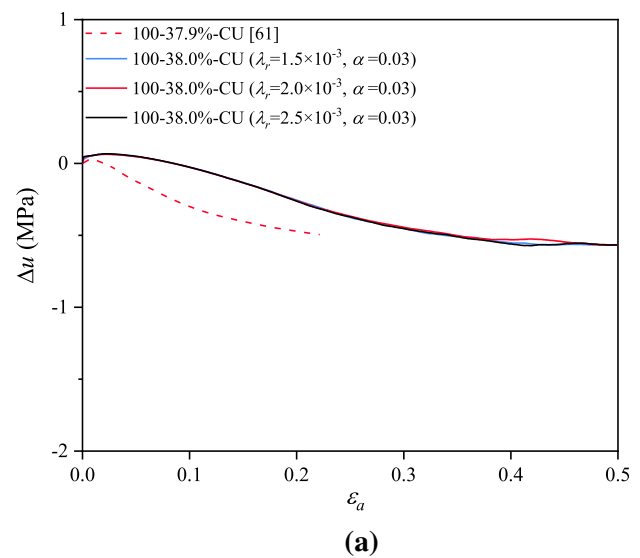
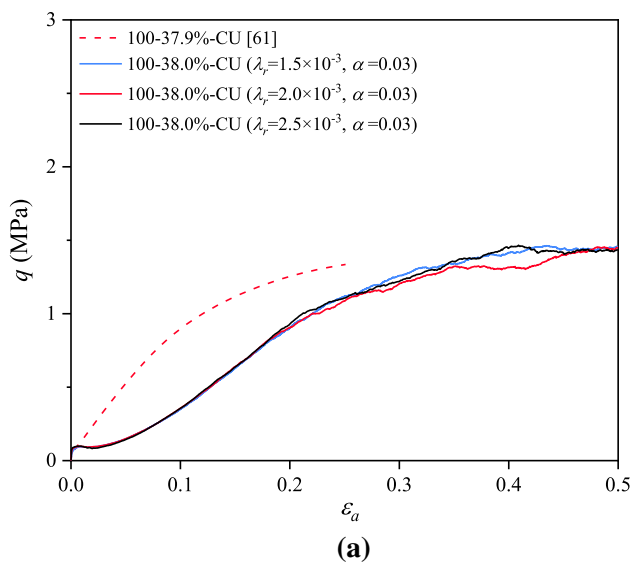


Fig. 15 Influences of **a** λ_r and **b** α on the deviatoric stress under undrained shearing

Fig. 16 Influences of **a** λ_r and **b** α on the excess pore water pressure under undrained shearing

between simulated behaviors and experimental data in undrained shearing. More studies still need to be carried out to further improve the performance of the proposed model.

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Declarations

Conflict of interest We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence the position presented in the manuscript entitled.

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